Hands-On vs. High-Tech:
How AI is Reshaping Physical Therapy Practices

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2025-2026

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12 May 2026

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Abstract

In today's modern age, many aspiring professionals question how safe their future job is from AI replacement. The research process described in this paper outlines how AI will impact physical therapists in the future. The research begins with medical publications, research articles about AI, machine learning, and rehabilitative tools to build a basic understanding of the field. Professionals were then inquired to participate in an interview. Professors, researchers, authors, therapists, and students all shared their insights, and one common pattern emerged. AI is an effective tool in boosting efficiency and holds promise in virtual contact or movement tracking technologies, but it lacks clinical decision making skills and empathy for patients, raising ethical concerns.

Literature Review

Last year, Bill Gates made waves in the clinical world when he appeared on "The Tonight Show", claiming "within 10 years, AI will replace many doctors and teachers—humans won't be needed 'for most things'". AI models have proven to be extremely beneficial in taking notes, handling medical records, and predicting patient outcomes. Models including Noterro have been adopted for tracking medical records. Several tools and apps like Kaia Health and Physitrack have been developed further by AI to guide patients in exercises virtually through their mobile device. On the other hand, many professionals have concerns over ethics and biases of AI, and patients' lack of trust may hinder the implementation of AI. While the mystery of AI's capability to assist or threaten the roles of human therapists remains ongoing, this paper examines all the factors involved. Although artificial intelligence is being increasingly used in physical therapy for movement analysis and increasing efficiency, it cannot replace clinical judgment or ethical decision making, making human therapist oversight essential for safe and effective patient rehabilitation.

The rise of AI in the physical therapy profession began gradually, with AI at first only performing administrative functions, such as documentation and charting medical records. Over the course of time, AI has moved towards movement analysis, rehabilitation, and the prediction of patient recovery. The benefits of the advancement of these functions can offer tremendous value to both the patient and the therapist. Such benefits include increased efficiency and greater access to therapy and regulation for exercise programs. As one of the many benefits of AI, there has been an accuracy score of 0.81 in tracking patient motion while on stroke rehabilitation machines, with pose estimation also keeping track of patient balance and position with impressive accuracy (Roggio, 2024). While many forms of AI must have its effectiveness studied and determined, additional concern has been placed over the bias and loss of human judgement that could be introduced in clinical settings.

However, the effectiveness of the use of AI varies. Most AI tracking tools have been shown to struggle with complex movements and most clinics don't have the resources for most solutions of that kind (Kapica, 2024). AI can find a bad stretch or form but may have trouble with complicated corrections because AI cannot read into the physical needs of the patient's pain. For instance, AI is helpful to spot a bad squat form but cannot adjust a patient's exercise protocol due to fatigue or anxiety (Alshami et al., 2025). Though there is proof that with the use of guided exercise by an AI it can reduce pain 30% more than standard therapy (Areias et al., 2024). Its use of documentation also is a benefit by reducing tedious tasks for the clinician, which gives them more time to interact with the patient as SOAP note time can be reduced by 50-70% (Net Health, 2024). Overall, machine learning programs can increase strength through patient results like reduced pain more so than standard therapy when the machine learning programs supplement therapist observation.

AI is greatly changing the role that physical therapists hold in therapy regarding how clinical decisions are made and how therapists interact with technology on a daily basis. Though AI systems can help clinicians make diagnoses and suggest a plan of treatment, experts say that AI should be seen as a helper, not a primary source of diagnoses. Therapists are responsible for the choices made in therapy under both a legal and ethical framework and will have to analyze the AI's findings before proceeding (AI in Therapy, 2025). Even though the information gained by an AI system regarding movement analysis or rehabilitation can be helpful in treatment, the therapist has to confirm the accuracy of the information to ensure it is safe for the individual. In healthcare, the individual working directly with the patient must take ultimate responsibility for how treatment decisions are made (Aliferis & Simon, 2024).

Beyond legal obligations, several therapists view holding onto their own autonomy over a case important for the profession. Surveys conducted on physical therapist attitudes toward AI in therapy concluded that most believe they should have ultimate control over how a case is handled. Concerns were repeatedly mentioned that if a therapist overrelied on an automated system there would be a decline in critical reasoning. A therapist could become too reliant on AI if the AI assumes primary responsibility of a patient's records, diminishing the ability for the therapist's skills to be gained from practice. Other researchers believe there is concern over the potential long term impacts of AI technology on therapists and their expertise in the profession (Physical Therapists' Perceptions and Attitudes Toward Artificial Intelligence in Healthcare and Rehabilitation, 2025). These fears underscore the need for technology's aid while human capabilities expand, rather than become reliant.

Rather than deeming the role of the physical therapist obsolete, AI has simply changed how they work. For many the repetitive task of documentation like SOAP notes is something that AI systems can and have been used for already. This change will allow for more direct interaction with the patient and the manual aspects of therapy. According to research done on the use of AI in physical therapy, therapists are starting to view the use of artificial intelligence as a way to aid them and not take away from the overall patient treatment (University of St. Augustine for Health Sciences, 2021). As a result, physical therapist roles may turn more toward a role that is responsible for monitoring the results that AI gives them and interpreting the data and relating it to their own clinical judgment and treatment plan.

Additionally, AI can help with accessibility. For people who are physically limited or are in remote locations where travel to a physical therapy facility may not be an option, remote therapy can allow patients to get treatment at home. The exercise program would be viewed through the smart phone or through another sort of device, with the patient seeing what is being expected in the video feed, and can follow their body closely and receive corrected feedback through the AI if they are not performing correctly. The use of artificial intelligence in rehabilitation machines will allow people that live far away from a clinic to extend the practice from the therapist into the home environment (Silva & Costa, 2023). It will also allow patients with transportation limitations and patients with difficulty with limited mobility and access to the use of physical therapy machines.

Yet, the accessibility gained by artificial intelligence may not be accessible by all in all of the United States. Facilities are required to have certain technology components available for all sensors, movement monitoring, and certain software programs. It may be more of a struggle for facilities that are more spread out and not as well funded. Most AI devices will also have trouble tracking more difficult and spontaneous movements, meaning it is unable to replace a therapist entirely. Thus, although AI is able to increase the availability of physical therapy, it will not work in all settings (Kapica, 2024).

While a large number of articles proved useful in showing the role of AI systems and rehabilitation technology, further original research was necessary to analyze the role of the rapidly increasing implementation of AI in the physical therapy profession through the viewpoint of current physical therapists working within the field. Professional literature is mainly focused on technical capacity, but more information is needed about the feelings and perceptions therapists, students, instructors, and researchers working in the field have on the use of AI in the field. Because of this gap in the research, interviews were conducted with physical therapists, researchers within sports medicine and rehabilitation science as well as AI researchers. These interviews helped to facilitate further research in that the findings shed light on the perceptions of the implications of AI within healthcare.

Methods

Primary research for this study was conducted through qualitative interviews with five individuals involved in physical therapy, rehabilitation, sports medicine, and healthcare research. Participants were selected based on their professional or academic involvement with rehabilitation sciences, physical therapy education, or AI technology. The sample included licensed physical therapists, a graduate student pursuing a Doctor of Physical Therapy degree, a university professor, a chiropractic and sports medicine specialist, and a researcher familiar with artificial intelligence systems.

The interviews were conducted virtually due to scheduling and geographic limitations. Participants were asked multiple open-ended questions concerning the real world implementation of artificial intelligence into physical therapy. These included questions regarding movement analysis, AI-powered documentation systems, electronic medical records, SOAP notes, predictive recovery models, clinical reasoning, bias in machine learning systems, access to care, and the future of artificial intelligence in the rehab setting. Questions were altered slightly based on the individual's area of professional expertise, and their answers were transcribed, summarized and coded for themes among participants.

Results

Several themes were consistent across all interviews, and many professionals reported similar beliefs. Almost all of the professionals believed that AI would continue to be increasingly useful in terms of its applications for administrative aspects of physical therapy (documentation, SOAP notes, billing, and electronic medical records). Dr. Plotkin believed that AI has already helped healthcare providers write reports, letters of medical necessity, treatment notes, and other paperwork. Dr. Ford had the same perspective that AI could save therapists a lot of time by aiding with documentation and allowing clinicians to interact more with patients.

Participants also consistently emphasized that AI cannot replace the "human element" of physical therapy. Dr. Ford explained that her profession has a lot of direct, hands-on care where therapists use manual treatment techniques that she believes cannot be replicated by AI. Ms. DeSola emphasized the need for empathy, communication, and awareness of emotional cues for patients working through complex medical or financial situations. Many participants said that clinical reasoning, the ability to judge a situation, and to communicate effectively remain human abilities. In regards to movement analysis and predictive models, several professionals reported using AI. Mr. Mariano described how AI could analyze motion data from wearables, motion-capture systems, and smartphone sensors for the purpose of identifying meaningful patterns regarding patient pain, fatigue, posture, and progress of recovery. Dr. Khan had similar reports that AI could also aid in outcome prediction, remote rehabilitation, and analyzing movements within a motion analysis lab.

Conversely, concerns regarding data collection and accuracy were repeatedly noted. Dr. Khan stated that rehabilitation datasets are difficult to acquire because of the many different variable conditions which require highly specific measurements of outcomes. Mr. Mariano stressed that patients will differ significantly based on the age of the patient, type of injury, patient motivation, and patient disability and access to healthcare. Several professionals also noted concerns about AI bias based on what data is available to machine learning systems. The overall consensus from many of the interviews was that AI should be used as an assistive technology rather than as a stand-alone, autonomous physical therapist. Every professional agreed that AI can greatly help with aspects of physical therapy and that humans can learn a lot from machines; but the majority of participants said that humans must still always review and ultimately be responsible for all AI generated inputs.

Discussion

The results of both the literature review and the interviews both supported the initial hypothesis, that while AI can augment and improve physical therapy, it cannot replace the human physical therapist. Throughout both secondary and primary literature reviews, one theme consistently recurred, AI should be used as a supportive, assistive technology instead of an autonomous one.

The interviews both confirmed and added to many of the ideas mentioned in the literature review. Researchers like Dr. Khan and Mr. Mariano emphasized many of the data-analyzing, predictive, and motion analysis features of AI for monitoring patient recovery in a virtual setting. Their views were not inconsistent with the machine learning and data prediction ideas within the literature. Practicing therapists such as Dr. Ford and Dr. Plotkin focused on the hands-on and patient-centric portions of therapy that can’t be reproduced with technology, clinical reasoning and manual treatment.

Although participants were diverse and from various professional backgrounds, there were a lot of shared sentiments amongst them. Researchers more highly emphasized the more predictive models and motion analysis systems, whereas practitioners seemed to prioritize automation of documentation, along with the undeniable value of hands-on treatment. However, all professionals universally believed that AI should be an assistive, but not autonomous, technology. Another prevalent theme that was agreed upon by many individuals was the role of documentation, which nearly everyone believed is one of the more tedious aspects of practice. From the literature review and interview results, there seems to be potential to reduce a lot of the time spent on documentation by implementing AI systems, allowing for more time spent with the patients.

Also highlighted within the interviews was concerns regarding the reliability and bias of AI technology. The warning was repeated multiple times that AI relies greatly on the data it is trained on, and without adequate diversity within data, outcomes from an AI may not be reliable or equitable across patient populations. This is again, in sync with the concerns surrounding the ethics of AI and healthcare algorithms.

This study does have a few limitations. This study only contained five interview subjects, a relatively small sample size. The study participants came from various fields of the rehabilitation sciences, but not all of the main fields of rehabilitation were covered in this study. Some of the subjects had a lot of experience using AI tools, whereas other subjects had very little exposure to any AI in practice. Due to limitations of interview format, it was difficult to conduct interviews over text/email and receive responses that were as thorough as they might have been in a face-to-face interview setting.

Despite these limitations, it provided very useful information as to the perception of AI in physical therapy as perceived by practicing professionals in the field. It would be useful for future studies to increase the number of subjects interviewed or have participants from a wider variety of fields of the healthcare professions or have actual patients discuss their comfort level with rehabilitation assisted by AI. Additional studies should also be conducted to analyze long-term patient outcomes with the assistance of AI-integrated rehab systems.

Overall, evidence shows that the integration of artificial intelligence into physical therapy will benefit the profession, but the human therapist will never be obsolete. Future studies demonstrate that intelligent machines may have an increased role in motion analysis, remote rehab, documentation systems, and predicting the outcome of rehab, however, the clinician will always have a necessary role to play in ethical decision making, clinical reasoning, hands-on care, and offering emotional support. The interviewees agreed that AI can and will improve practice efficiency and patient outcomes by adding value to the system but that it should not replace humans in the long run. The integration into physical therapy will focus more on the aspects of what AI can do as a supporting tool and the challenge will be implementing technology into practice in a way that maintains the necessary human touch within rehabilitation practices.

Conclusion

Artificial intelligence has the potential to significantly transform numerous aspects of the physical therapy profession. Machine learning technologies allow for an increased analysis of movement, the automation of documentation, and an improvement in how therapists manage large volumes of patient data. These improvements to the system could lead to greater clinic efficiency and better access to rehabilitation through the use of remote systems and computer-guided exercise routines. Despite these potential gains, the data within this paper clearly supports that artificial intelligence cannot ever replace the integral human side of physical therapy. Hands-on treatment, clinical reasoning and communication are essential components of providing high-quality physical therapy care. The interview subjects all seemed to confirm the idea that AI will function as an assist to therapists and not a stand-alone source of help for patients, as it is essential that therapists maintain a therapeutic relationship with their patients. It is foreseeable that a blended model will evolve between the human and machine with therapists providing clinical judgment, manual therapy skills and emotional support while AI handles much of the more basic tasks. The real challenge for physical therapy will not be whether AI can replace therapists, but how the profession can effectively integrate technology while still maintaining the human connection that rehabilitation depends on.

Acknowledgements

I would like to thank the professionals who advised me along the way. Their perspectives and experiences provided valuable insight into the evolving role of artificial intelligence within physical therapy and rehabilitation sciences.

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Appendices

Appendix A: Sample Interview Questions

What are the most promising areas of research involving AI in physical therapy or rehabilitation science today?

How has AI changed the way researchers analyze human movement and rehabilitation outcomes?

Many AI systems rely on large datasets of patient movement and recovery outcomes. What challenges exist when collecting or training AI models using rehabilitation data?

How reliable are current AI-based movement analysis tools compared to traditional assessment by trained therapists?

What limitations currently prevent AI from replacing clinical reasoning performed by physical therapists?

How has AI improved documentation systems such as SOAP notes or electronic medical records in physical therapy practice?

Some researchers have raised concerns about bias in medical AI systems. How might bias affect AI tools used in rehabilitation settings?

What role should physical therapists play in guiding the ethical use of AI technologies in healthcare?

How do you personally envision AI influencing physical therapy over the next decade?

Appendix B: Interview Participants

Dr. Fayaz Khan – Professor and rehabilitation researcher

Marciano Mariano – Healthcare researcher and AI policy advisor

Dr. Larry Plotkin – Chiropractor and sports medicine specialist

Dr. Tami Ford – Licensed physical therapist

Dani DeSola – Doctor of Physical Therapy graduate student